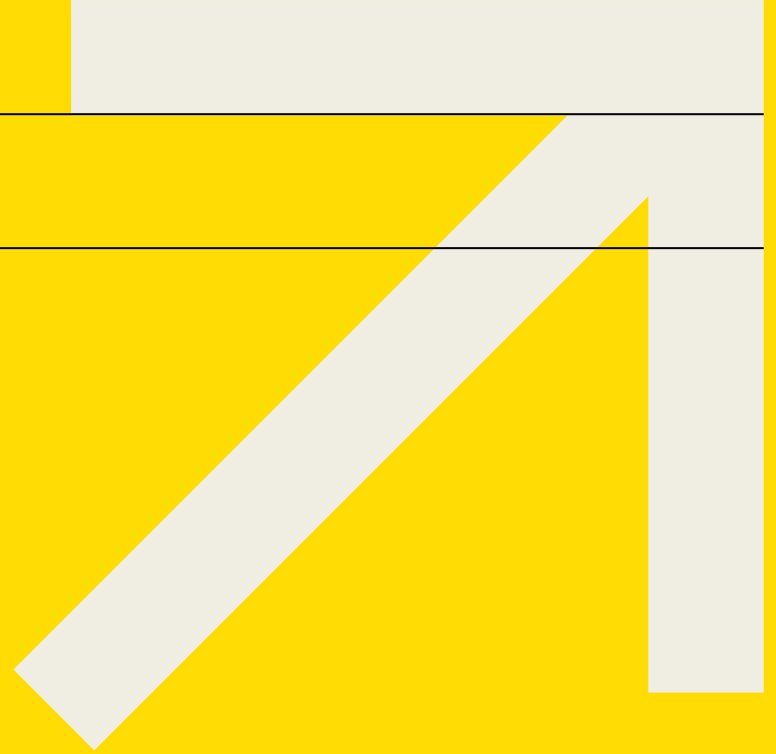


# Family Decisions: A game



# The Game purpose



The game tracks **Family Harmony**, **Stress Levels**, and **Schedule Conflicts** while teaching real-world scheduling optimization through the universal challenge of managing a busy family life!

## What are we trying to do:

- Optimize parent schedule by dividing to them the tasks
- Manage fixed schedules vs flexible tasks; priorities and needs vs wants
- Account for activities that must happen in sequence
- Balance family harmony, efficiency, and stress levels
- Avoiding scheduling overlaps and resource conflicts



# Objective



We need to schedule the daily activities of the family Pickleshaw and their busy schedule. See different possible solutions and how they change the overall goals - achieve **Family Harmony**, reduce **Stress Levels**, and solve any **Schedule Conflicts**

# The game



## How to play:

- 1) Assign how is doing the activity (mom or dad)
- 2) Press "Schedule activity" then "Advance time"
  - The activity will be assign for the first available spot of the person assigned for it
- 3) Select another task and repeat previous steps until all tasks are assigned to the schedule of the Pickleshaw family.

# Questions to think

## Questions

What happens with you assign more things for one parent than the other?

Some solution might be optimal for one objective but not for others.

Can you get a solution that is 100% optimal for all 3 objectives? How does that relate to your life?

What changed in the solution? How is it balanced? Does it reflect how you feel or not? Why do you think that is?

## Test it:

Give all the assignments to one parent and then just the rest for the second parent.

Give the same amount of work for both parents.

Try it!

Choose who is doing each of the activities as you would in your life

# Operations Research and decision-making

This game demonstrates core **Operations Research (OR)** principles through realistic family scheduling challenges, transforming abstract mathematical concepts into tangible decision-making scenarios.

Each parent assignment involves **decision analysis**: evaluating alternatives, assessing risks (stress levels), and predicting outcomes. The real-time feedback loop teaches how decisions impact system performance—a key OR methodology.

The scoring system (efficiency, family harmony, stress levels) provides quantitative feedback typical of OR solutions, enabling continuous improvement and strategy refinement.

Scheduling Problem is a classic problem in the Operations Research field. The gamification of this classic problem makes abstract OR concepts accessible, demonstrating how mathematical optimization principles solve real-world scheduling challenges in any organization or system with limited resources and competing demands.



## Operations Research Concepts worked in this game:

- **Resource Allocation:** Optimizing parent assignments
- **Constraint Satisfaction:** Managing fixed schedules vs flexible tasks
- **Dependency Management:** Activities that must happen in sequence
- **Multi-Objective Optimization:** Balancing family harmony, efficiency, and stress levels
- **Conflict Resolution:** Avoiding scheduling overlaps and resource conflicts
- **Critical Path Analysis (CPM):** The dependency chain (breakfast → school transport → work availability) illustrates **CPM** concepts.

Thank you!

And hope to see you soon!